

- Statement

For  $m \times m$  matrix  $A, A_n$ , if  $A, A_n$  is invertible and  $A_n = A + a$  with  $a_{ij} = o_p(1)$ ,  $1 \leq i, j \leq m$ , then  $\|A_n^{-1} - A^{-1}\|_{op} = o_p(1)$ .

- proof

$$A_n^{-1} = (A + a)^{-1} = A^{-1}(I + A^{-1}a)^{-1} = A^{-1} \left( I + \sum_{l=1}^{\infty} (-A^{-1}a)^l \right)$$

Then

$$\|A_n^{-1} - A^{-1}\|_{op} = \|A^{-1} \sum_{l=1}^{\infty} (-A^{-1}a)^l\|_{op} \leq \|A^{-1}\|_{op} \left\| \sum_{l=1}^{\infty} (-A^{-1}a)^l \right\|_{op}$$

Using  $\|A^{-1}a\|_{op} = o_p(1)$ ,

$$\leq \|A^{-1}\|_{op} \frac{\|A^{-1}a\|_{op}}{1 - \|A^{-1}a\|_{op}} = o_p(1)$$

Claim that  $\|A^{-1}a\|_{op} = o_p(1)$

$[A^{-1}a]_{ij} = \sum_k (A^{-1})_{ik} a_{kj} = \sum_k (A^{-1})_{ik} o_p(1) = o_p(1)$  for  $1 \leq i, j \leq m$ .

$$\begin{aligned} \sup_{x_1^2 + \dots + x_n^2 \leq 1} \|A^{-1}a\mathbf{x}\| &= \sup_{x_1^2 + \dots + x_n^2 \leq 1} \sqrt{\sum_{i=1}^m \left( \sum_{j=1}^m ([A^{-1}a]_{ij} x_j) \right)^2} \\ &\leq \sqrt{\sum_{i=1}^m \left( \sum_{j=1}^m ([A^{-1}a]_{ij})^2 \right)} = o_p(1) \end{aligned}$$

# Frobenius, operator norm

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$$\|A\|_{op} = \sup_{\|x\| \leq 1} \|Ax\|, \|A\|_F = \sqrt{\sum a_{ij}^2} = \sqrt{\text{tr}(A^T A)}$$

Using  $A^T A = U D U^T$  with  $U U^T = U^T U = I$ ,  $D = \text{diag}(\lambda_i)$ , singular values  $\lambda_i \geq 0$  of  $A$

$$\|A\|_{op}^2 = \sup_{\|x\| \leq 1} x^T A^T A x = \sup_{\|x\| \leq 1} x^T U D U^T x = \sup_{\|y\| \leq 1} y^T D y = \lambda_{\max}$$

Now with  $\text{rank}(A^T A) = r$ ,

$$\sqrt{\lambda_{\max}} \leq \sqrt{\sum \lambda_i} \leq \sqrt{r \lambda_{\max}}$$

$$\|A\|_{op} \leq \|A\|_F \leq \sqrt{r} \|A\|_{op}$$